Think like a Bayesian

Restaurant stars and well water

Work in pairs. Keep a phone calculator handy.

One idea for today

Combine what you **knew before** (the **prior**) with what the **data** say. The more data you have, the more you trust them. The result is your **best guess** (the **posterior**).

Exercise 1. Which restaurant?

8 minutes

A typical restaurant in town gets about **4.0 stars**. That is our **prior**. We say it is worth **10 reviews**.

1. Restaurant A has 2 reviews, average 5.0. Restaurant B has 800 reviews, average 4.7. Which one would you pick? ☐ A ☐ B

2. Fill in the table. Restaurant A is done for you.

$$\text{trust } w = \frac{\text{reviews}}{\text{reviews} + 10}, \qquad \text{best guess} = w \times \text{average} + (1 - w) \times 4.0$$

Restaurant	Reviews	Average	Trust w	Best guess
A	2	5.0	$2/12 = 0.17$	$0.17 \times 5.0 + 0.83 \times 4.0 = 4.17$
B	800	4.7	_____	_____
D	1	1.0	_____	_____

3. A dropped from 5.0 to 4.17. B hardly moved. Why?

4. D has one 1-star review, yet its best guess is 3.73. Why?

Exercise 2. Is the well water safe? *10 minutes*

A family tests its well for arsenic **ten times**. The EPA limit is **10 μg/L**. Question: **how often is the water above the limit?**

Prior: in nearby wells, 5 of 10 past tests were above the limit.

> **Keep score.** Start with **Above = 5** and **Below = 5** (the nearby wells). After each test, add 1 to Above or to Below.
>
> **Best guess** = $\dfrac{\text{Above}}{\text{Above} + \text{Below}}$ (2 decimals)

5. Fill in the score card. The first rows are done for you.

Test	Arsenic (μg/L)	Above or below 10?	Above	Below	Best guess
—	*nearby wells (prior)*		5	5	0.50
1	14	above	6	5	0.55
2	7	below	6	6	0.50
3	12	_____	____	____	_____
4	18	_____	____	____	_____
5	11	_____	____	____	_____
6	15	_____	____	____	_____
7	9	_____	____	____	_____
8	13	_____	____	____	_____
9	14	_____	____	____	_____
10	14	_____	____	____	_____

6. The well's own tests alone: ____ of 10 above, a share of _____. The nearby wells: 0.50. Your final best guess: _____.

Where does it fall? _____

7. Same answer with the restaurant formula. The prior is worth 10 tests and the family has 10 tests, so $w = 10/(10 + 10) = 0.5$:

$$0.5 \times 0.80 + 0.5 \times 0.50 = _____$$

8. A second family tested only **twice**, both above the limit. Their score: Above = ____, Below = ____, best guess = _____.

Which family's guess stays closer to the neighbours' 0.50? _____ Why?

Exit ticket *2 minutes*

Sullivan County expects 7 lung cancer cases; Philadelphia expects 1,219. Both SIRs are pulled toward the state average. Which one is pulled more?

☐ Sullivan ☐ Philadelphia